Machine learning-based synthetic CPT prediction in offshore windfarms with sparse data

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Summary

With the rising demand for offshore wind energy, there is an increased need to acquire Cone Penetration Test (CPT) data more rapidly. Approaches for predicting synthetic CPTs from geophysical data using machine learning have developed. A good performance of the machine learning often requires a substantial dataset, which is not always available, particularly for windfarms in their initial development phases characterized by limited CPT data.

This study investigates the capability of the machine learning approach for predicting synthetic CPTs with limited data in an offshore windfarm in the German North Sea. The results show that with limited training data, this approach can generate realistic smoothed synthetic cone resistance for most locations situated outside and above the glacial valleys. However, challenges arise within glacial valleys due to complex sub-surface conditions, seismic data limitations, inadequate training data. To enhance predictions, future work may benefit from increasing training data and considering segregating training and prediction exercise based on geology.

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Introduction

The Cone Penetration Test (CPT) is a geotechnical investigation method used to determine the soil's properties and assess its mechanical behaviour. Commonly measured parameters include cone resistance (qc), sleeve friction(fs) and pore water pressure(u2). These parameters are essential for understanding the sub-surface conditions and guiding the foundation design for the offshore wind turbines. In recent years, with the rising demand for offshore wind energy, there's an increased urgency to acquire CPT data more rapidly. Consequently, methodologies for deriving synthetic CPTs using geostatistics or machine learning techniques have been developed (Sauvin et al., 2019; Chen et al., 2021; Sauvin et al., 2022; Siemann et al., 2022). Both methodologies utilize seismic data, albeit to varying extents. The geostatistics-based approach mainly employs horizons interpreted from seismic data to guide the property modelling. In contrast, the machine learning-based approach leverages seismic data throughout the entire interval of interest. It establishes a relationship between the CPT and seismic data, enabling the prediction of synthetic CPTs at every seismic trace location.

A good performance of the machine learning often requires substantial amount of data. Previous studies have demonstrated the potential of machine learning in predicting synthetic CPTs for offshore wind farms, particularly when abundant CPT data is available (NGI, 2022). However, a large dataset may not always be available, especially for small sites or offshore windfarms in their early stages of development when only limited CPTs have collected. This leads to pertinent questions: How well can this approach perform in offshore windfarms with sparse CPT data? And will its performance vary across geological conditions of differing complexities? In this study, the capability of the machine learning approach in predicting cone resistance for an offshore windfarm site with limited CPT data is explored. Furthermore, its strengths and limitations are analysed in various geological conditions, both within and outside of glacial valleys.

Data and study area

The data utilized for this study are from the offshore windfarm area N-7-2 located in the German North Sea, including geotechnical, geophysical and geological data. The dataset comprises Cone Penetration Tests and P-S logging from 10 locations (as shown in Figure 1a), 2D Multi-Channel Seismic (MCS) data (with locations denoted by the green lines in Figure 1a), and interpreted stratigraphic unit boundaries (such as the pink line in Figure 1b). The depth interval penetrated by CPTs are down to 80 meters below the seafloor.

The sequence observed in this area is a sandy sedimentary succession spanning from the early Pleistocene to the Holocene, strongly influenced by the Pleistocene glaciations resulting in large-scale structures such as buried glacial valleys. One glacial valley goes in a N-S direction in the westernmost part of the area, and some are in the eastern part of the area (Figure 1). Some of the CPTs are drilled in the buried valleys where seismic data display low amplitude, multiples, and discontinuous or chaotic patterns (Figure 1; BSH, 2022).

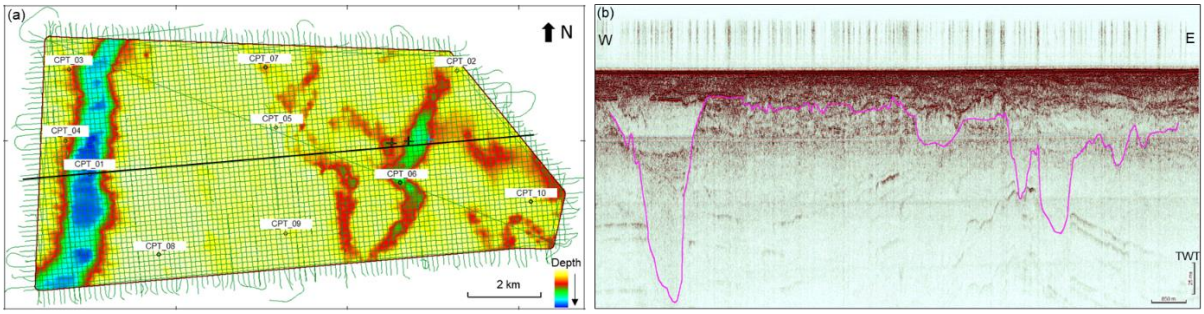


Figure 1 *(a) Map view showing the locations of the Cone Penetration Test data and the glacial valleys*

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in the study area. (b) A W-E Multi-Channel Seismic profile crossing the glacial valleys; the pink line outlines the interpreted base of the glacial sediments.

Approach

A dataset including CPT parameters, TWT, co-ordinates, geological soil unit and co-located seismic-derived parameters is compiled. The CPT data is converted to time domain to match with the seismic data. Given the differing sampling rate of CPT and seismic data, one can either upscale the CPT data to align with the seismic data or downscale the seismic data to match the CPT data; the latter can be beneficial for small datasets as it increases the number of instances available for training. In this study, both ways are explored.

Deep Neural Network (DNN) is used to predict cone resistance. The input variables include acoustic impedance derived from the 2D Multichannel seismic profiles through a post-stack seismic inversion process, envelop which is a type of seismic attribute representing the magnitude of the seismic reflections at different time intervals, geological soil unit interpreted from the 2D seismic data, and Two-Way Time (TWT). The selection of input variables and model hyperparameters is based on a trial-and-error process. Leave-One-Out Cross-Validation is used to evaluate the performance of the trained model at different CPT locations.

Result

Figure 2 shows blind cone resistance prediction at each CPT location, plotted against the actual cone resistance data. The prediction at each location is made by a neural network trained with data from the other 9 locations. It shows that the predictions are quite consistent with the data at locations CPT_2, CPT_5, CPT_7, CPT_8, CPT_9, and CPT_10. Those locations are mostly outside of the glacial valleys (Figure 1). A significant discrepancy between predictions and data was observed at locations CPT_1, CPT_3, CPT_4, and CPT_6, especially in intervals below 80-90 milliseconds (roughly 25-30 meters beneath the seafloor). All these locations fall within the glacial valleys during the mentioned intervals. Overall, the predicted cone resistance in the first 25-30 meters below the seafloor appears realistic in all locations.

In this particular case with sparse data, the trained neural networks are not capable of predicting the fine-scale variations, especially in the deeper intervals. The neural networks trained with upscaled data, where CPT data is averaged to match the seismic data, tend to generate more smoothed synthetic qc profiles (illustrated by the grey curves in Figure 2). Conversely, those trained with downscaled data, where seismic data is interpolated to match with the finer sampling rate of CPT data, can capture more fine-scale variations (as shown by the orange curves in Figure 2).

The unsatisfactory predictions within the glacial valleys are believed to result from several factors. Firstly, there is a high variation in cone resistance within these valleys. Secondly, the seismic data in those valleys present challenges: they display low amplitude, multiples, and discontinuous to chaotic reflection patterns. The seismic signatures in addition vary between the valley in the west and the ones in the east (BSH, 2022). Lastly, the available training data is insufficient to train a model that can accurately capture the intricate relationship between the varied cone resistance and the complex seismic facies.

To improve predictions within the glacial valleys, two strategies are suggested. First, augmenting the training data from these valleys. Second, training and predicting separately for areas inside and outside the valleys. As cone resistance and seismic signatures vary distinctively between these areas, such segregation is anticipated to reduce heterogeneity, simplifying the relationship between cone resistance and seismic data.

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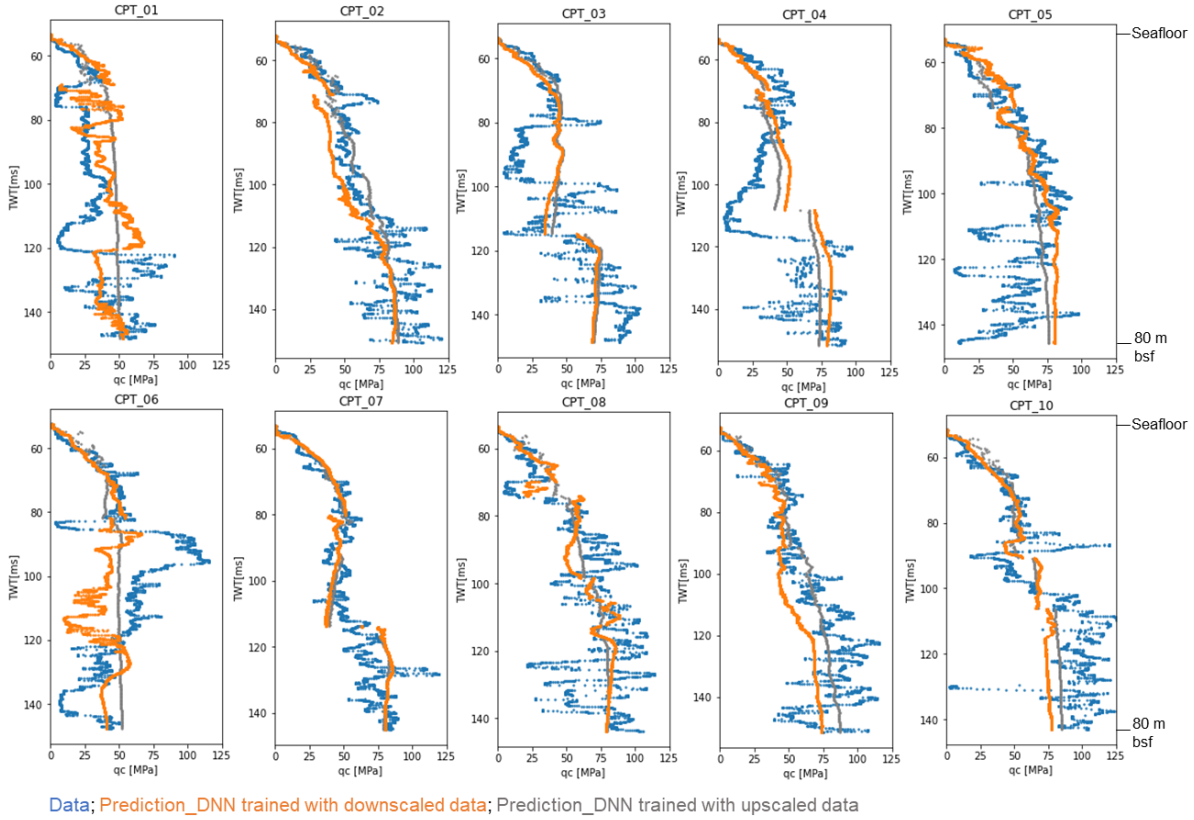


Figure 2 *Cone resistance (qc) prediction at each CPT location; Predicted qc in each location is made by a deep neural network trained with data from the other 9 locations.*

Conclusions

In this study, the capability of machine learning approach in predicting cone resistance at sites with sparse CPT data has been investigated. The findings highlight the strength of this approach, particularly when applied to areas outside and above the glacial valleys.

Within the glacial valleys, significant discrepancies between predictions and data were observed. These discrepancies likely arise from the inherently complex and varied nature of the sub-surface in these areas, combined with challenges posed by the seismic data and a lack of sufficient training data.

To enhance predictions, future work may benefit from increasing training data and segregating training and prediction exercise based on geology, particularly differentiating between glacial valley and non-glacial valley areas.

Acknowledgements

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References

BSH [2022] Geological Report of the Preliminary Investigation of FEP-Site N-7.2 – Investigation Area: N-07-02. Federal Maritime Agency of Germany, Hamburg.

Chen, J., Vissinga, M., Shen, Y., Hu, S., Beal, E. and Newlin, J. [2021] Machine learning–based digital integration of geotechnical and ultrahigh–frequency geophysical data for offshore site characterizations. *Journal of Geotechnical and Geoenvironmental Engineering*, *147*(12), 1-14.

NGI [2022] Ten noorden van de Waddeneilanden Wind Farm Zone Integrated Ground Model. Doc No. 20190798-04-R. Client RVO.

Siemann, L., Morales, N., Chautru, J.M., Pein, T. [2022] Prediction and probability estimation of geotechnical parameters for offshore windfarm site characterization using geostatistical simulation. *EAGE GET Conference, Extended Abstracts,* p.1-5.

Sauvin, G., Vanneste, M., Vardy, M.E., Klinkvort, R.T. and Carl Fredrik, F. [2019] Machine learning and quantitative ground models for improving offshore wind site characterization. *Offshore Technology Conference*, Houston, TX, pp.18

Sauvin, G., Vardy, M., Klinkvort, R.T., Vanneste, M., Forsberg, C.F. and Kort, A. [2022] State-of-the-art ground model development for offshore renewables–TNW case study. *EAGE GET Conference, Extended abstracts,* p.1-5.

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